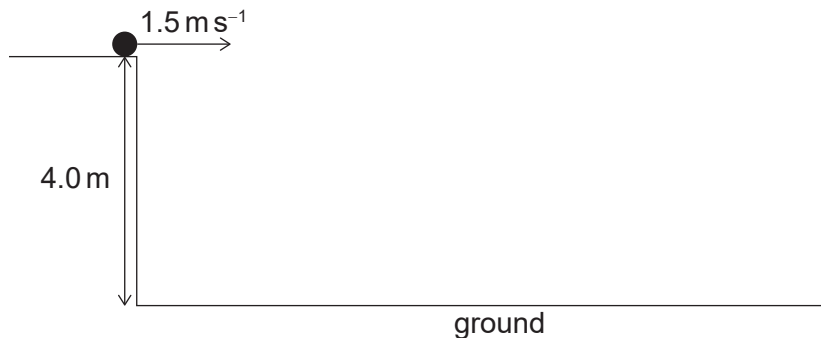


PAPER 1 / QUESTION 27

28 April 2022 / TZ1 / HL Paper 1 / original question 5 / 2222-6507

27. A stone is kicked horizontally at a speed of 1.5 m s^{-1} from the edge of a cliff on one of Jupiter's moons. It hits the ground 2.0 s later. The height of the cliff is 4.0 m . Air resistance is negligible.

What is the magnitude of the displacement of the stone?



- A. 7.0 m
- B. 5.0 m
- C. 4.0 m
- D. 3.0 m